

Practice quiz week 1

Here is your first practice quiz. The quiz is not timed and you have unlimited attempts to take it.

This is not required, and your score will not be incorporated into your final grade.

1 Multiple Choice 0.5 points

Scientists observed that frogs with toe pads live in trees, not in water. They stated, "toe pads are important for life in trees." This statement is an example of a(n)...

- ☐ hypothesis
- ☒ prediction
- ☐ theory
- ☐ law
- ☐ experiment



2

Multiple Choice 0.5 points

Suppose the researchers locate a colony of coral established in a natural setting and begin collecting data on the percentage bleaching in the colony at regular time intervals. They observe that the percentage bleaching increases just after the water temperatures are observed to rise dramatically. How would this investigation be classified?

- ☒ Observational study
- ☐ Controlled experiment
- ☐ Variable experiment
- ☐ Predictive study

3

Multiple Choice 0.5 points

All of the field mice, grasshoppers, deer, hawks, plants, and other organisms in a particular valley in Ohio constitute a(n)...

- ☒ community
- ☐ abiotic system
- ☐ population
- ☐ landscape
- ☐ biosphere

4

Multiple Choice 0.5 points

Suppose scientists discover a planet very similar to Earth. What information will be important in determining the growing seasons in different regions of this new planet?

- ☒ Temperature and precipitation
- ☐ Temperature only
- ☐ Precipitation only
- ☐ Soil type only
- ☐ Precipitation and soil type

5

Multiple Choice 0.5 points

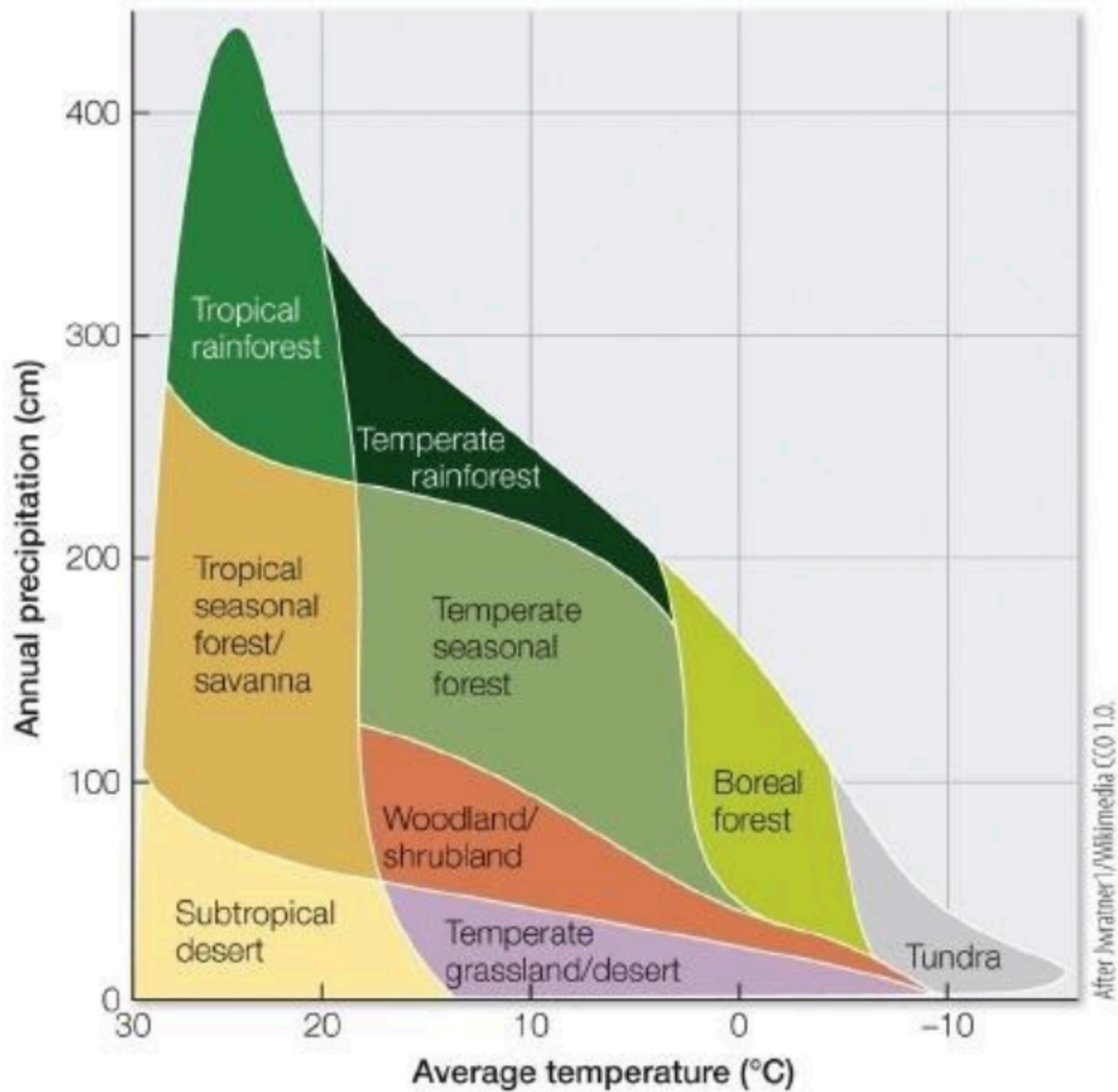
The coast of California and the coasts of Mediterranean countries have similar climates with warm, dry summers and mild, rainy winters. They are inhabited by ecologically similar organisms. These two regions can thus be said to be in the same...

- ☒ biome
- ☐ geological zone
- ☐ gyre
- ☐ photic zone
- ☐ biotic interchange

6

Multiple Choice 1 point

Refer to the figure.



If, over a long period of time, a region that is currently in the temperate seasonal forest biome becomes considerably colder, this region will most likely become...

- ☐ boreal forest
- ☐ temperate rain forest
- ☐ tundra
- ☐ tropical rain forest
- ☐ woodland or shrubland

7

Multiple Answer 1 point

A researcher hypothesizes that forest fire is an important factor in determining whether a particular grassland ecosystem in northern California is maintained as a grassland rather than a woodland. Which of the following would be a scientific way to test her hypothesis? Check all that apply.

- ☐ A. Burn the whole ecosystem and record what happens after.
- ☐ B. Compare grass and tree cover in 20 climatically similar locations in Northern California that have had different amounts of time lapse between the last fire.
- ☐ C. Divide the ecosystem into several small plots, burn half of them and leave the other half as controls. Record the change in vegetation for several years after the two treatments.
- ☐ D. Compare a burned ecosystem in California with a non-burned ecosystem in North Carolina
- ☐ Set up plots that are fenced to exclude deer and control plots of the same size that are not fenced. Compare the vegetation after 1 year.

8

Multiple Choice 0.5 points

Which of these is not an example of climate?

- ☐ The average temperature of College Park in July?
- ☐ The mean rainfall of Seattle
- ☐ A 70% chance of rain tomorrow
- ☐ The length of the winter in Nome, Alaska

9

Multiple Choice 1 point

The per capita growth rate of a population of 200 seals is 0.05. A correct interpretation of this is that:

- ☐ There is less than one birth per female season a year
- ☐ The survival rate per seal is 95%
- ☐ The population is expected to grow by 10 seals this year.
- ☐ The population is expected to decline by 5% this year.
- ☐ The population is expected to decline by 95% this year.

10

Multiple Choice 1 point

The Lake Titicaca from (*Telmatobius culeus*), also called the 'Scrotum Frog' for its soft saggy skin is currently endangered. Last year researchers studying a population of 200 frogs in a nature reserve in Boliva recorded 84 new frogs and 80 deaths. What is the *per capita* population growth rate?



- ☐ 204
- ☐ 4
- ☐ 0.4
- ☐ 0.02

11

Multiple Choice 1 point

A small population consisting of 50 Edith's checkerspot butterflies is declining ($r = -0.02$). Assuming no butterflies emigrate, how many immigrants does the population need to prevent decline?

- ☒ 1
- ☐ 2
- ☐ 5
- ☐ 20
- ☐ impossible to tell